



Cassiterite

△ **Cassiterite rough**, also known as tin oxide

Cassiterite, a major source of tin, takes its name from *kassiteros*, the Greek word for the element. The vast majority of cassiterite is opaque black or brown, but occasionally transparent, reddish-brown crystals are found and can be faceted for collectors. Facet-grade crystals are sometimes recovered from rock, but most gemstone material is collected from stream gravels where it has weathered from the rocks in which it formed. Cassiterite continues to be mined for tin, especially in Malaysia, Thailand, Indonesia, and Bolivia.

Specification

Chemical name	Tin oxide	Formula	SnO_2	Colours	Medium to dark brown
SG	6.7–7.1	RI	2.0–2.1	Hardness	6–7
Lustre	Adamantine to metallic				
Streak	White, greyish, brownish				
Locations	Portugal, Italy, France, Czech Republic, Brazil, Myanmar				



Cassiterite in matrix | Rough | Here, finely formed cassiterite crystals rest on a groundmass of muscovite. Their shiny lustre is adamantine (diamond-like).



Gemmy crystals | Rough | Transparent, reddish-brown, gem-quality crystals of cassiterite rest on this groundmass of massive cassiterite.



Cassiterite crystals | Rough | This cluster of sharp, well-defined, dark crystals with excellent lustre grow outwards from a lump of massive cassiterite.

Tin from cassiterite

Bronze Age to baked beans

Cassiterite-sourced tin has been traded across the Mediterranean world since the Bronze Age began in about 3000BCE. Tin is the essential component (along with copper) of bronze, and the dark colour of cassiterite makes it easy to see against the granite in which it typically forms. In modern times, the plating of steel with non-toxic and corrosion-resistant tin has revolutionized food storage with the creation of the tin can, often just called a “tin”.



Tin cans Tin derived from cassiterite has been essential in the long-term preservation of food.



Faceted oval | Colour variety | This brilliant-cut oval shows the yellow-brown colour typical of most faceted cassiterites. Its colour flickers in the light like a diamond.



Faceted round | Colour variety | Here, the dark inclusions contrast with the colourless cassiterite, an unusual colour considered rare enough to facet into a cut gemstone.